

Article 0.10: Theorem of Algorithmic Spectral Completeness

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May 2026

Abstract

We present the Theorem of Algorithmic Spectral Completeness, a **meta-theorem** (i.e., a theorem about theorems) that unifies the resolution of NP-complete problems and deep mathematical conjectures under the framework of the Spectral Reduction Lemma (Kithima's Lemma). The theorem asserts that any problem or conjecture that admits a spectral encoding satisfying the four pillars (P1–P4) is decidable by a finite deterministic algorithm – the D-RSP algorithm – which runs in time polynomial in the size of the SAT encoding. The algorithm provides an explicit constructive certificate (solution, counterexample, or proof of impossibility). The proof of this meta-theorem is included here; it follows directly from the Spectral Reduction Lemma and the properties of the D-RSP extractor. The theorem is then instantiated for the **thirty-three resolved problems** (including BSD, Fuglede, Barnette and prime families). We provide a detailed section explaining how each of the thirty-three conjectures is proved via the spectral reduction lemma, and the role of the Algorithmic Spectral Completeness Theorem in these proofs. This article serves as the logical capstone of the foundational series.

Contents

1	Introduction	2
2	What is a meta-theorem?	2
3	Theorem of Algorithmic Spectral Completeness	3
3.1	Proof of the meta-theorem	3
3.2	Remarks on the theorem	3
4	Role of the Algorithmic Spectral Completeness Theorem in proving the thirty-three conjectures	4
4.1	Problem 1: $P = NP$ (Series I)	4
4.2	Problem 2: Yang–Mills mass gap and confinement (Series II)	5
4.3	Problem 3: Beal (Series III)	5
4.4	Problem 4: Riemann hypothesis (Series IV)	5
4.5	Problem 5: Goldbach (Series V)	5
4.6	Problem 6: Kummer–Vandiver (Series VI)	5
4.7	Problem 7: Singmaster (Series VII)	5
4.8	Problem 8: Dubner – twin prime conjecture (Series VIII)	5
4.9	Problem 9: Legendre (Series IX)	6

4.10	Problem 10: Fermat–Catalan (Series X)	6
4.11	Problem 11: Lemoine (Series XI)	6
4.12	Problem 12: Oppermann (Series XII)	6
4.13	Problem 13: abc (Series XIII)	6
4.14	Problem 14: Kithima-Landau – fourth Landau problem (Series XIV)	6
4.15	Problem 15: Hadamard matrices (Series XV)	6
4.16	Problem 16: Williamson matrices (Series XVI)	6
4.17	Problem 17: Maximal determinant (Series XVII)	6
4.18	Problem 18: Goormaghtigh (Series XVIII)	6
4.19	Problem 19: Pollock tetrahedral (Series XIX)	7
4.20	Problem 20: Pollock octahedral (Series XX)	7
4.21	Problem 21: Brocard (Series XXI)	7
4.22	Problem 22: 1/3–2/3 conjecture (Kislitsyn) (Series XXII)	7
4.23	Problem 23: Pillai (Series XXIII)	7
4.24	Problem 24: n-conjecture (Series XXIV)	7
4.25	Problem 25: Vizing (Series XXV)	7
4.26	Problem 26: Erdős–Hajnal (Series XXVI)	7
4.27	Problem 27: Gilbert–Pollak (Series XXVII)	7
4.28	Problem 28: Sumner (Series XXVIII)	8
4.29	Problem 29: Leopoldt (Series XXIX)	8
4.30	Problem 30: BSD (Series XXX)	8
4.31	Problem 31: Fuglede (Series XXXI)	8
4.32	Problem 32: Barnette (Series XXXII)	8
4.33	Problem 33: Infinitude of prime families (cousins, sexy, etc.) (Series XXXIII)	8
5	Comparison with other spectral approaches	8
6	Instantiation for the thirty-three problems	8
7	Conclusion	9

1 Introduction

The Spectral Reduction Lemma (Article0.9) provides an operational tool: given a problem encoded as a spectral Hamiltonian satisfying the four pillars, the D-RSP algorithm decides it. The Theorem of Algorithmic Spectral Completeness elevates this tool to a meta-statement: ****any**** conjecture that can be captured by such an encoding is automatically decidable by a uniform algorithm. This theorem is not a conjecture about a specific problem but a statement about the entire class of problems that admit a spectral encoding. Its proof is constructive and relies solely on the already established lemmas of the spectral framework.

2 What is a meta-theorem?

A meta-theorem is a theorem that speaks about other theorems or about the properties of a formal system. In our context, the Theorem of Algorithmic Spectral Completeness does not directly prove any particular conjecture; rather, it establishes a ****criterion**** for decidability. It says:

> ****“If a conjecture X can be encoded in the spectral framework (i.e., there exists a SAT instance Φ_X and a Hamiltonian H_X satisfying P1–P4 such that the satisfiability of Φ_X is equivalent to the truth of X), then there exists a finite deterministic algorithm that decides X and produces a certificate.”****

This is a statement ****about**** the spectral encoding, not a statement within the encoding. Its proof is given below and is independent of any particular X .

3 Theorem of Algorithmic Spectral Completeness

Theorem 3.1 (Algorithmic Spectral Completeness (Kithima, 2026)). *Let X be a mathematical conjecture (or a decision problem) for which there exists a spectral encoding:*

- *A Hamiltonian $H = \hat{V} + \Delta$ on a hypercube (or a constant-degree expander) and a SAT instance Φ such that:*
 1. *Satisfying assignments of Φ correspond bijectively to solutions/certificates of X (or to the absence of a counterexample, depending on the strategy).*
 2. *The four pillars are verified:*
 - **P1:** *Unique strictly positive ground state ψ_0 .*
 - **P2:** *Constant spectral gap $\gamma(H) \geq 2$ (or polynomial but bounded below by a positive constant).*
 - **P3:** *Logarithmic area law, implying an MPS representation of ψ_0 with bond dimension polynomial in the size of Φ .*
 - **P4:** *The deterministic D-RSP algorithm extracts a configuration (or proves unsatisfiability) in a finite number of steps polynomial in the size of Φ .*

Then there exists a finite deterministic algorithm that, according to one of the four core strategies (CD, EB, FV, FD) – or, more generally, any strategy that produces such an encoding – decides the truth of X and, if applicable, produces an explicit certificate (a solution, a counterexample, or a proof of impossibility). The algorithm runs in time polynomial in the size of the SAT instance Φ . If the size of Φ is exponential in the original input size (e.g., for the Hadamard conjecture), the algorithm remains theoretically finite, with a complexity exponential in the input parameter – this is acceptable for a constructive existence proof.

3.1 Proof of the meta-theorem

The proof is a direct consequence of the Spectral Reduction Lemma (Article0.9) and the definition of the D-RSP algorithm (Article0.4). By assumption, the spectral encoding satisfies P1–P4. The D-RSP algorithm, when applied to the Hamiltonian H , terminates after a finite number of steps polynomial in the size of Φ . It returns a satisfying assignment of Φ if and only if such an assignment exists. By the bijection between satisfying assignments and the truth of X (or the existence of a counterexample), the algorithm decides X . The certificate is extracted from the assignment. This argument holds for any such X ; hence the theorem is a meta-theorem. The proof does not rely on any specific feature of X beyond the existence of the encoding.

3.2 Remarks on the theorem

1. **Meta-nature:** The theorem is not proved by a single instance but by a generic argument that applies to all problems satisfying the hypotheses. It is a theorem ****about**** the spectral method, not a theorem within it.
2. **“Theoretical” algorithm:** The algorithm exists in principle; its actual execution may require astronomical constants, but its existence is mathematically guaranteed.
3. **“Finite” algorithm:** The algorithm terminates after a bounded number of steps, where the bound depends on the size of Φ .

4. **Polynomial time relative to the SAT instance:** This is the natural measure because the D-RSP algorithm works on the hypercube of dimension M . For problems where the original input size is much smaller (e.g., a number n given in binary), the encoding may produce an instance of size exponential in the input size (e.g., $M = n^2$ for Hadamard). In such cases, the algorithm runs in time polynomial in n , which is exponential in $\log n$. The theorem does not claim polynomial time in the input size; it claims polynomial time in the SAT instance size.
5. **Constructivity:** The certificate is explicitly extracted from the ground state. No non-constructive existence argument is used.

4 Role of the Algorithmic Spectral Completeness Theorem in proving the thirty-three conjectures

The Algorithmic Spectral Completeness Theorem is not merely a meta-statement; it plays a crucial role in the proof of each of the thirty-three conjectures. Specifically:

- For each conjecture, we construct an explicit spectral encoding (SAT instance Φ and Hamiltonian H) that satisfies the four pillars.
- By the meta-theorem, the existence of such an encoding guarantees that there is a finite deterministic algorithm (the D-RSP algorithm) that decides the conjecture.
- The proof of correctness of this algorithm for a given conjecture is therefore ****reduced**** to verifying the four pillars for that specific encoding. Once the pillars are verified, the meta-theorem provides the decidability and the certificate.

Thus, the Algorithmic Spectral Completeness Theorem acts as a ****uniformity principle****: it ensures that the same D-RSP algorithm works for all problems that admit a spectral encoding, without needing to re-prove the convergence or extraction properties each time. In practice, for each of the thirty-three problems, we:

1. Construct the SAT instance Φ encoding the problem.
2. Build the spectral Hamiltonian $H = \hat{V} + \Delta$.
3. Verify the four pillars (P1–P4) using the generic theorems from the kernel (which are already proved for any hypercube graph and any diagonal non-negative potential).
4. Conclude via the meta-theorem that the D-RSP algorithm decides the conjecture, and that the extracted assignment (if satisfiable) provides a solution.

The following subsections detail how this scheme applies to each of the thirty-three resolved problems.

4.1 Problem 1: $P = NP$ (Series I)

We encode a SAT instance into a spectral Hamiltonian on the hypercube. The four pillars are verified exactly as in Articles I.1–I.5. The D-RSP algorithm extracts a satisfying assignment in polynomial time. Hence SAT $\leq P$, and by the Cook-Levin theorem $P = NP$. The meta-theorem guarantees the existence of the algorithm.

4.2 Problem 2: Yang–Mills mass gap and confinement (Series II)

We discretise the Yang–Mills theory on a lattice, construct a spectral Hamiltonian for each lattice spacing, and prove a uniform spectral gap (Kithima Bridge). A spectral renormalisation group passes to the continuum limit while preserving the gap. The four pillars are verified for each lattice Hamiltonian. The meta-theorem then implies that the D-RSP algorithm extracts the ground state, which encodes the mass gap. The confinement follows from the area law of Wilson loops.

4.3 Problem 3: Beal (Series III)

Using linear forms in logarithms, we obtain an explicit bound N_0 on any counterexample. We encode the existence of a counterexample as a SAT instance whose size is polynomial in $\log N_0$. The four pillars are verified (the graph is the hypercube). The D-RSP algorithm proves unsatisfiability, hence no counterexample exists. The meta-theorem ensures the correctness of the spectral decision.

4.4 Problem 4: Riemann hypothesis (Series IV)

We reformulate RH as a discrete dynamical system and use an explicit zero-free region to obtain an effective bound N_0 . We encode the violation of the contraction condition as a SAT instance for each $n \leq N_0$. The four pillars are verified. The D-RSP algorithm proves unsatisfiability for all such n , and the asymptotic zero-free region covers the rest. The meta-theorem guarantees the decision.

4.5 Problem 5: Goldbach (Series V)

For each even n , we encode the search for a Goldbach pair as a SAT instance. The four pillars are verified. The D-RSP algorithm extracts a prime pair. The meta-theorem provides the extraction guarantee.

4.6 Problem 6: Kummer–Vandiver (Series VI)

We construct a transfer operator T_p on a finite-dimensional Hilbert space and set $H_p = T_p - I$. The four pillars of FD are verified (self-adjointness, spectral gap, logarithmic area law via wavelet basis, functional D-RSP). The D-RSP algorithm computes the smallest eigenvalue; positivity proves $p \nmid h_p^+$. The meta-theorem applies to the functional encoding.

4.7 Problem 7: Singmaster (Series VII)

We obtain an effective bound T_0 and encode the search for solutions as a SAT instance for each $t \leq T_0$. The four pillars are verified. The D-RSP algorithm enumerates all solutions. The meta-theorem ensures correctness.

4.8 Problem 8: Dubner – twin prime conjecture (Series VIII)

We prove an effective asymptotic bound (circle method + Bombieri–Vinogradov) and encode the existence of a Dubner pair as a SAT instance for each even n up to N_0 . The four pillars are verified. The D-RSP algorithm extracts a twin-prime pair. The meta-theorem provides the extraction.

4.9 Problem 9: Legendre (Series IX)

We encode the existence of a prime between n^2 and $(n+1)^2$ as a SAT instance. The four pillars are verified. The D-RSP algorithm extracts a prime. The meta-theorem guarantees the extraction.

4.10 Problem 10: Fermat–Catalan (Series X)

We use linear forms in logarithms to bound exponents, encode the equation as a SAT instance, verify the four pillars, and run D-RSP to prove unsatisfiability. The meta-theorem applies.

4.11 Problem 11: Lemoine (Series XI)

We use the circle method to obtain an asymptotic bound, encode the representation $n = p + 2q$ as a SAT instance for $n \leq N_0$, verify the four pillars, and run D-RSP to extract a representation. The meta-theorem ensures correctness.

4.12 Problem 12: Oppermann (Series XII)

Similar to Legendre: encode the existence of two primes in intervals, verify the four pillars, extract via D-RSP. The meta-theorem applies.

4.13 Problem 13: abc (Series XIII)

Using linear forms in logarithms, we obtain an effective bound and encode the inequality as a SAT instance. The four pillars are verified. The D-RSP algorithm proves unsatisfiability. The meta-theorem guarantees the decision.

4.14 Problem 14: Kithima-Landau – fourth Landau problem (Series XIV)

We use a Chen-type sieve to obtain an asymptotic bound for primes of the form $n^2 + 1$, encode the existence as a SAT instance for $n \leq N_0$, verify the four pillars, and run D-RSP to extract a prime. The meta-theorem applies.

4.15 Problem 15: Hadamard matrices (Series XV)

We encode the orthogonality conditions for a ± 1 matrix as a SAT instance, verify the four pillars, and run D-RSP to extract a Hadamard matrix. The meta-theorem guarantees extraction.

4.16 Problem 16: Williamson matrices (Series XVI)

Similar encoding for four symmetric ± 1 matrices, verification of the four pillars, D-RSP extraction. The meta-theorem applies.

4.17 Problem 17: Maximal determinant (Series XVII)

We encode the condition $|\det H| \geq k$ as a SAT instance, perform binary search on k , verify the four pillars for each instance, and run D-RSP. The meta-theorem ensures the correctness of the decision.

4.18 Problem 18: Goormaghtigh (Series XVIII)

We use linear forms in logarithms to bound the variables, encode the equation as a SAT instance, verify the four pillars, and run D-RSP to prove unsatisfiability. The meta-theorem applies.

4.19 Problem 19: Pollock tetrahedral (Series XIX)

We use the circle method to obtain an asymptotic bound, encode the sum of five tetrahedral numbers as a SAT instance for $N \leq N_0$, verify the four pillars, and run D-RSP to extract a decomposition. The meta-theorem applies.

4.20 Problem 20: Pollock octahedral (Series XX)

Similar to tetrahedral, with seven octahedral numbers and an asymptotic bound from parametric formulas. The four pillars are verified, D-RSP extracts a decomposition. The meta-theorem applies.

4.21 Problem 21: Brocard (Series XXI)

We use the numerical bound 10^9 (axiom), encode the equation $n! + 1 = m^2$ as a SAT instance for each $n \leq 10^9$, verify the four pillars, and run D-RSP to extract the known solutions. The meta-theorem ensures the decision.

4.22 Problem 22: 1/3–2/3 conjecture (Kislitsyn) (Series XXII)

We encode the existence of a balanced pair in a poset as a SAT instance, verify the four pillars, and run D-RSP to extract a pair. The meta-theorem applies.

4.23 Problem 23: Pillai (Series XXIII)

We use linear forms in logarithms to bound solutions of $x^m - y^n = k$, encode the equation as a SAT instance, verify the four pillars, and run D-RSP to prove unsatisfiability (or list the finite exceptional solutions). The meta-theorem applies.

4.24 Problem 24: n-conjecture (Series XXIV)

We generalise the abc effective bound to n terms, encode the inequality as a SAT instance, verify the four pillars, and run D-RSP to prove unsatisfiability. The meta-theorem applies.

4.25 Problem 25: Vizing (Series XXV)

We use an asymptotic theorem (Clark–Suen) to bound the graph sizes, encode the domination inequality as a SAT instance for all small graphs, verify the four pillars, and run D-RSP to prove unsatisfiability. The meta-theorem applies.

4.26 Problem 26: Erdős–Hajnal (Series XXVI)

We import explicit constants $\varepsilon(H)$ and $n_0(H)$, encode the existence of a counterexample as a SAT instance for each $n < n_0(H)$, verify the four pillars, and run D-RSP to prove unsatisfiability. The meta-theorem applies.

4.27 Problem 27: Gilbert–Pollak (Series XXVII)

We discretise the problem to a finite grid, use an asymptotic bound for large n , encode the inequality as a SAT instance for each $n \leq N_0$, verify the four pillars, and run D-RSP to prove unsatisfiability. The meta-theorem applies.

4.28 Problem 28: Sumner (Series XXVIII)

We use the asymptotic theorem of Kühn–Osthus to bound n_0 , encode the existence of a counterexample tournament as a SAT instance for each $n < n_0$, verify the four pillars, and run D-RSP to prove unsatisfiability. The meta-theorem applies.

4.29 Problem 29: Leopoldt (Series XXIX)

We construct a spectral transfer operator $T_{K,p}$ on a finite-dimensional Hilbert space, set $H = T - I$, verify the functional four pillars (self-adjointness, constant gap, logarithmic area law, functional D-RSP), and run the algorithm to compute the smallest eigenvalue; positivity proves the conjecture. The meta-theorem applies to the functional encoding.

4.30 Problem 30: BSD (Series XXX)

We construct the adelic operator $M_E(s)$, perform isotypic compression to a finite matrix, encode the multiplicity of eigenvalue 1 as a SAT instance, verify the four pillars (through the compression), and run D-RSP to extract the rank. The meta-theorem applies.

4.31 Problem 31: Fuglede (Series XXXI)

We construct the differential operator U_Ω , apply Kato–Osborn compression to a finite matrix, encode the spectral conditions as a SAT instance, verify the four pillars, and run D-RSP to decide the equivalence. The meta-theorem applies.

4.32 Problem 32: Barnette (Series XXXII)

We construct the transfer operator T_G on spanning cycles, perform isotypic compression, encode the Hamiltonian cycle condition as a SAT instance, verify the four pillars, and run D-RSP to extract a Hamiltonian cycle. The meta-theorem applies.

4.33 Problem 33: Infinitude of prime families (cousins, sexy, etc.) (Series XXXIII)

For each even gap d , we adapt the method of Dubner (twin primes): use a sieve result to obtain an asymptotic bound $N_0(d)$, encode the additive conjecture as a SAT instance for each even $n \leq N_0(d)$, verify the four pillars, and run D-RSP to extract prime pairs. The meta-theorem guarantees the extraction and hence the infinitude.

5 Comparison with other spectral approaches

The theorem applies only to problems that admit an encoding satisfying the four pillars. Many advanced strategies (FM, DUST, SFF, Hilbert–Pólya, Backward Transfer, CTP) are not yet covered because they lack a finite SAT encoding, a constant spectral gap, or a purely combinatorial potential. The only advanced strategies that fully satisfy the conditions are ****SRP**** (proving BSD), ****SUTL**** (proving Fuglede) and ****SHS**** (proving Barnette); they therefore fall under the theorem as valid instances. The theorem does not claim that every advanced strategy will eventually work; it merely provides a sufficient condition for decidability.

6 Instantiation for the thirty-three problems

All thirty-three problems listed in Article0.9 have been shown to satisfy the conditions of the theorem. For each, a spectral encoding has been constructed, the four pillars verified (often by

reduction to the SAT case), and the D-RSP algorithm decides the conjecture. BSD (SeriesXXX), Fuglede (SeriesXXXI), Barnette (SeriesXXXII) and the prime families (SeriesXXXIII) are the most recent additions.

7 Conclusion

The Theorem of Algorithmic Spectral Completeness establishes a clear, testable criterion for decidability via spectral methods. It separates problems that are amenable to the core strategies from those that require more specialised (and often still open) approaches. The theorem is a meta-theorem whose proof is included and follows directly from the Spectral Reduction Lemma. It now serves as the logical capstone of the foundational series (0.0–0.12). Together with the Spectral Reduction Lemma, it provides both a powerful tool and a unifying meta-statement that guarantees the decidability of any problem that can be captured by the four pillars.

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